Contents

Laws and Regularities Classification

Paper 2C: The Coordination Collapse Framework

Epistemic Status Classification for Each Relationship

Overview

This document classifies the coordination collapse relationships into three tiers based on their epistemic status:

1. **Core Laws** - Strong theoretical derivation + solid empirical support
2. **Supporting Regularities** - Empirical patterns with theoretical grounding but more uncertainty

3. **Theoretical Extensions** - Frontier research, speculative but mathematically coherent

This hierarchy is deliberate: it invites empirical and theoretical work to upgrade regularities to laws, rather than presenting all relationships as equally settled.

Classification Summary

ID	Name	Type	Evidence	Equation
1	Threshold	**Core Law**	Strong	$\theta = C/(B+C) \approx 0.375$
2	Cascade Dynamics	**Core Law**	Strong	$dH_3/dt = -\lambda_1(\theta-H_3) - \lambda_2(\theta-H_3)^2$
9	Positive Feedback	**Core Law**	Strong	$d^2H_3/dt^2 > 0$ when $H_3 > \theta$
10	Percolation	**Core Law**	Strong	$\theta \approx p\_c(z=4) \approx 0.388$
5	Recovery	**Derived Law**	Moderate	$P(\text{recovery}) \approx 0.15 \times e^{(-0.05\times \text{years})}$
3	Network/Redundancy	Regularity	Moderate	$v\_cascade \propto 1/\sqrt{R}$
4	Modernization	Regularity	Moderate	$\lambda = \lambda\_base \times F\_info \times F\_econ \times F\_network$
6	Visibility (Masked Trust)	Regularity	Weak	$H_3\_apparent = H_3\_true + H_3\_masked(R)$
7	Intervention ROI	Regularity	Moderate	ROI ≈ 10× above θ vs 1/10 below
8	Dark Trust	Regularity	Weak	$H_3\_total = H_3\_light + H_3\_dark$
11	Learning ($\mu \approx 0$)	Regularity	Moderate	$\theta(t) \approx \theta_0$ over 5000 years
12	Glass Ceiling	Regularity	Moderate	$K\_max \approx 0.85$
13	Information Hub	Extension	Theoretical	$I(H_3; H_{-3}) = \max\_i\ I(H\_i; H_{-i})$
14	Phase Transition	Extension	Theoretical	$F(m,T) = a(T)m^2 + bm^4$; $\beta=½$
15	Metacognitive	Extension	Theoretical	$M = K \times R\_model \times T\_response$
16	Evolutionary Stability	Extension	Theoretical	$dp/dt = p(1-p)(-c) < 0$
17	AI Coordination Paradox	Extension	Theoretical	$dH_3/dt\_AI = dH_3/dt(1 + A\_pos)$

Tier 1: Core Laws

These have clean mathematical derivations AND solid empirical support.

Law 1: Coordination Threshold

Statement: There exists a critical trust threshold $\theta \approx 0.375$ below which defection becomes the Nash equilibrium and coordination collapse becomes highly probable.

Derivation: Game-theoretic from Prisoner's Dilemma: $\theta = C/(B+C)$ where B=benefit, C=cost of cooperation.

Evidence Status: **Strong** - LOOCV validation on 35 historical cases: 94% predictive accuracy - Independent derivations converge: PD (0.375), half-entropy (0.382), percolation (0.388) - See MAIN_PAPER Figure 2, SI Table S3

Note on θ value: The value $\theta \approx 0.375$ is model-dependent, not a fundamental constant. Different game payoff structures yield different thresholds. The empirical estimate $\theta\_emp \approx 0.375 \pm 0.025$ is consistent with multiple theoretical derivations. The proximity to $1/\varphi^2 \approx 0.382$ is noted as a mathematical curiosity, not claimed as evidence.

Law 2: Cascade Dynamics

Statement: Below threshold, trust decline accelerates quadratically:

$$dH_3/dt = -\lambda_1(\theta - H_3) - \lambda_2(\theta - H_3)^2$$

Derivation: From contagion dynamics with positive feedback below threshold.

Evidence Status: **Strong** - $R^2 \approx 0.74$ across 35 historical collapse cases - Calibrated $\lambda_1 \approx 0.015$, $\lambda_2 \approx 0.08$ (per decade) - See MAIN_PAPER Figure 3, SI Section 4.2

Law 9: Positive Feedback

Statement: Above θ, cooperation reinforces itself (virtuous cycle); below θ, defection reinforces itself (vicious cycle).

$$\text{If } H_3 > \theta: d^2H_3/dt^2 > 0 \text{ (recovery accelerates)}$$
$$\text{If } H_3 < \theta: d^2H_3/dt^2 < 0 \text{ (collapse accelerates)}$$

Derivation: Game-theoretic from repeated game dynamics and trust reciprocity.

Evidence Status: **Strong** - Supported by Ostrom (1990), Axelrod (1984), social capital literature - Observed in 28/35 historical cases - See MAIN_PAPER Section 5.1

Law 10: Percolation

Statement: The threshold θ corresponds to the percolation transition of the coordination network.

$$\theta \approx p_c(z=4) = (z-1)^{-1} \approx 0.333 \text{ to } 0.388$$

Derivation: Network percolation theory; coordination as path connectivity.

Evidence Status: **Strong** - Independent mathematical derivation yields $p_c \approx 0.388$ for $z=4$ networks - Overlaps with empirical $\theta_{emp} \approx 0.375$ within uncertainty - See SI Section 6.1, Figure S12

Law 5: Recovery (Derived)

Statement: Recovery probability decays exponentially with time below threshold.

$$P(\text{recovery} \mid H_3 < \theta) \approx 0.15 \times e^{(-0.05 \times \text{years\_below\_}\theta)}$$

Derivation: Hazard model fitted to recovery attempts in historical data.

Evidence Status: **Moderate** - Based on 12 recovery attempts in dataset - Binomial CI [0.08, 0.22] for base rate - See MAIN_PAPER Table 2, SI Section 5.3

Tier 2: Supporting Regularities

Empirical patterns with theoretical grounding but more uncertainty.

Regularity 3: Network/Redundancy

Statement: Cascade velocity inversely proportional to network redundancy.

$$v_{cascade} \propto \lambda(\theta - H_3)^2 / \sqrt{R}$$

Evidence: Rome ($R \approx 3$) vs Soviet Union ($R \approx 1.8$) collapse speed comparison.

Status: **Moderate** - Fits case studies, needs broader validation.

Regularity 4: Modernization Pressure

Statement: Modernization increases cascade speed through multiple channels.

$$\lambda = \lambda_{base} \times F_{info} \times F_{econ} \times F_{network}$$

Where F factors represent information speed, economic integration, and institutional coupling.

Evidence: $\lambda \approx 21\times$ pre-industrial for modern systems.

Status: **Moderate** - Predictive but difficult to falsify independently.

Regularity 6: Visibility (Masked Trust)

Statement: Observable trust can mask underlying fragility.

$$H_3\_apparent = H_3\_true + H_3\_masked(R)$$
$$H_3\_masked \leq 0.25 \text{ (bounded)}$$

Evidence: Pre-collapse optimism observed in multiple cases.

Status: **Weak** - Conceptually important but hard to operationalize.

Regularity 7: Intervention ROI Asymmetry

Statement: Intervention returns highly asymmetric around threshold.

$$ROI(H_3 > \theta) \approx 10\times \text{ to } 30\times$$
$$ROI(H_3 < \theta) \approx 0.1\times \text{ to } 0.3\times$$

Evidence: Comparative effectiveness analysis across interventions.

Status: **Moderate** - Policy-relevant, supported by case studies.

Regularity 8: Dark Trust

Statement: Total coordination capacity includes hidden reserves.

$$H_3\_total = H_3\_light + H_3\_dark$$

Where H_3_dark represents informal networks, shadow institutions, underground co-operation.

Evidence: Post-collapse resilience variations.

Status: **Weak** - Conceptually crucial but still being operationalized.

Regularity 11: Threshold Stability (μ ≈ 0)

Statement: Civilizations do not learn to lower θ over time.

$$d\theta/dt \approx \mu \approx 0 \text{ over historical record}$$

Evidence: θ stable across 5000 years of civilization.

Status: **Moderate** - Observational; absence of evidence vs evidence of absence.

Regularity 12: Glass Ceiling

Statement: Maximum sustainable K bounded below 1.

$$K_{max} \approx 0.85 \pm 0.05$$

Evidence: No civilization sustained $K > 0.9$ for extended periods.

Status: **Moderate** - Strong empirical bound, theoretical derivation in progress.

Tier 3: Theoretical Extensions

Frontier research—mathematically coherent but speculative.

Extension 13: Information Hub

Statement: H_3 (trust) has highest mutual information with other harmonies.

$$I(H_3; H_{-3}) = \max_i I(H_i; H_{-i})$$

And θ corresponds to half-maximum coordination entropy: $S_{coord}(\theta) = S_{max}/2$.

Status: **Theoretical** - Plausible framework, needs explicit derivation and data validation.

Extension 14: Phase Transition

Statement: Coordination collapse follows mean-field second-order transition.

$$F(m,T) = a(T)m^2 + bm^4$$
$$\beta = 1/2,\ \gamma = 1,\ \nu = 1/2 \text{ (mean-field exponents)}$$

Status: **Theoretical** - Elegant mapping but historical data too sparse to estimate exponents rigorously.

Extension 15: Metacognitive Collapse

Statement: Civilizations lose self-awareness before collapse.

$$M = K \times R_{model} \times T_{response}$$
$$\text{Collapse when } M < C_{env}$$

Status: **Theoretical** - Interesting framework for “why they can’t see it coming”; needs operationalization.

Extension 16: Evolutionary Stability

Statement: Under baseline replicator dynamics, cooperation is not ESS.

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dp/dt = p(1-p)(-c) < 0
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Scope limitation: This applies to one-shot, well-mixed populations without institutions. Does NOT contradict network reciprocity, repeated games, or group selection mechanisms.

Status: **Theoretical** - Clean math but claim must be scoped carefully.

Extension 17: AI Coordination Paradox

Statement: AI amplifies coordination dynamics in both directions.

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dH₃/dt_AI = dH₃/dt_nat × (1 + A_pos(t))
dλ/dt_AI = dλ/dt_nat × (1 + A_neg(t))
```

Status: **Theoretical** - Framework for analyzing AI impact; connects to Technology Evaluation Matrix.

How to Read This Classification

For Academic Reviewers

We distinguish between a small set of **Core Laws** (threshold, cascade, feedback, percolation) that we claim have strong support, and a larger set of **Supporting Regularities** that capture observed patterns but remain open to refinement.

This hierarchy: - Shows epistemic humility - Identifies which claims are grounded vs speculative - Invites empirical work to upgrade regularities to laws

For Practitioners

Core Laws provide hard constraints on what trajectories are sustainable.

Regularities provide guidance on mechanisms and interventions, with appropriate uncertainty.

Extensions provide conceptual frameworks for frontier issues (AI, cosmic implications).

Evidence Summary

Tier	Count	Average Evidence	Use
Core Laws	4 (+1 derived)	Strong	Hard constraints
Regularities	7	Moderate-Weak	Guidance with uncertainty
Extensions	5	Theoretical	Frontier frameworks

Citation

When referencing this classification:

```
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```

Status: Living document **Last Updated**: December 2025 **Purpose**: Distinguish epistemic status of framework components